

Response to Previous Review

This proposal is a resubmission of the proposal previously titled “**Designing agronomic and genetic strategies to maximize soybean systems’ performance under climate change**”.

We appreciate the comments from previous review and have considered them carefully. Major aspects are addressed in this letter and in more detail in our revised proposal. Structural changes include removing objective 1 and expanding objective 2 by adding a new task. The new task replaced the maturity group × plant density experiment with a rainfed × irrigated experiment and replaced measurements of greenhouse gas emissions with measurements of crop evapotranspiration. An on-farm experiment was added to quantify crop evapotranspiration in real field conditions and maximize stakeholder involvement. We ensured alignment with current Administration Policy and Federal Priorities. A focus was given to identifying strategies for current drought and high temperature events rather than climate change projections. Novelty aspects of the proposal were synthesized in one paragraph. We focused on the improvement of model capabilities to better capture physiology in simulations, which was indicated as a strength by most reviewers. New preliminary data was added to reinforce our systems-level approach. We added significant description of the growth chamber study and the APSIM calibration/validation process. More specifically:

Overall objective: “genetic strategies” sentence not adequate (reviewer 1) – replaced with “physiological strategies” to better represent proposal scope.

Approach: lacks detail (reviewer 1) – methodology of the field and growth chamber experiment expanded. One more replication added to the field experiment. Dynamic temperature setup up added to the growth chamber experiment, and light levels adjusted based on the growth chamber capabilities and relevant literature.

Preliminary data: perceived limited data (reviewer 1) – new preliminary data generated, and novelty aspects of previously presented data emphasized.

Central hypothesis: lack of novelty (reviewer 1) – reformulated to reflect the novel aspect of the proposal – improvement of capabilities in a dynamic crop growth model by including more physiological dynamics to better capture crop response to temperature and water (*see overview section*). Also, novel aspects were synthesized in a new paragraph (*see last paragraph in preliminary data section*).

Approach: APSIM calibration and validation lack detail (reviewer 2). Data split and methodology for calibration and validation described in more detail.

Approach: additional APSIM assessment (reviewer 2). An additional dataset from USDA-NASS was included to assess APSIM in reproducing historical crop yield variability for the study region.

Data management: no specific repositories or catalogs named other than GitHub (reviewer 2). We have expanded the data management plan and included repositories such as the USDA Ag Data Commons.

Approach: suggested addition of rainfed vs. irrigated field experiment (reviewer 3): experimental design modified and added as a new task (*see task 1.2*).

Approach: ambitious number of combinations in the growth chamber experiment (reviewer 3) – rearranged to ensure feasibility.

AGRONOMIC AND PHYSIOLOGICAL STRATEGIES TO MAXIMIZE SOYBEAN SYSTEMS' PERFORMANCE TO DROUGHT AND HIGH-TEMPERATURE RISKS

INTRODUCTION

Overview

Extreme variability in weather conditions, like drought and high temperatures, are emerging threats to US soybean production, but a comprehensive quantification of their long-term impacts on seed yield and system-level performance is lacking. Elli et al. (2022) found that these abiotic stressors can decrease soybean seed yields by 19% in the US if systems-specific adaptive strategies are not identified. To put that in perspective, soybean is a \$2.4 billion crop in Arkansas, and extreme weather implications could cost up to \$456 million to growers (Elli et al., 2022; Arkansas Agriculture Profile, 2023). Despite the growing magnitude of abiotic stressors limiting soybean yields in the US, complex crop interactions call for systems-level perspectives (Matthews and Burgess, 2024) to enhance understanding of how agronomic management and crop physiological traits interact under certain weather conditions. Lack of this understanding limits our ability to identify effective solutions for preventing soybean yield losses.

Existing crop growth models lack physiological detail to quantify the impacts of drought and high temperatures on crop growth and yields (Nóia Júnior et al., 2025), which is the essential first step to design assertive agronomic recommendations that maximize final yields and cropping systems' metrics like water use efficiency, seasonal crop evapotranspiration, and water losses like runoff and drainage (Freitas et al., 2025). Thus, there is a critical need to enhance model capabilities and leverage them within a cross-scale systems framework to identify solutions that can potentially maximize yields without increasing farmer input, like seasonal water demand. Without this framework, we will be unable to improve recommendations that provide a consistent yield increase for target environments.

Our long-term goal is to help stakeholders in US agriculture overcome weather challenges by better understanding the physiological mechanisms underlying crop yields in relation to the production system. The overall objective of this proposal is to identify agronomic and physiological strategies that will alleviate the negative impacts of weather variability and enhance soybean yield while maximizing water use efficiency in target environments using a system-level framework. Our central hypothesis is that a cross-scale and system-level understanding of soybean response to year-to-year weather variability will facilitate the conceptualization of solutions to maximize seed yields and reduce farmer input under drought and high temperature risks. Our preliminary results support this hypothesis by demonstrating that improved agronomic management and physiological traits provided opportunities to enhance soybean yield without increasing seasonal water demand for target weather scenarios.

The fundamental rationale of the proposed project is that once our crop physiology-modeling framework is created and tested, system-specific solutions can be identified that will provide actionable short-term (agronomic management) and long-term (physiological traits to potentially include in breeding programs) information to prevent soybean yield losses and design more efficient cropping systems for target weather conditions. Our focus will be on soybean production systems in the US Mid-South, but the approaches implemented here could serve as a model for similar research in other production systems and environments as appropriate. Our combined expertise in soybean physiology and modeling (Elli et al., 2022; Elli and Archontoulis,

2023) and monitoring environmental fluxes (Runkle et al., 2019; Reavis et al., 2024) makes us an ideal team to lead this project. Our specific objectives are to:

Objective #1: Implement new algorithms within a cropping system model (Agricultural Production Systems Simulator - APSIM) that operate at the leaf level and further test the model's ability to simulate soybean growth, seed yield, and water use. Working hypothesis – new algorithms that incorporate more realistic trait physiology will enhance model predictability and explanatory power of soybean physiological processes under variable weather conditions (Boote et al., 2021).

Objective #2: Identify the most effective management and physiological strategies for target environments in the context of year-to-year weather fluctuations that will increase soybean seed yields and water use efficiency. Working hypothesis – given the significant genotype, environment, and management interactions, dynamic and system-specific agronomic recommendations will maximize seed yields and water use efficiency (Elli et al., 2022).

Upon the completion of this project, we expect to have a crop growth model with enhanced physiological mechanisms and implement this model into a novel framework capable of generating short- and long-term recommendations to maximize the resiliency of soybean systems to year-to-year weather fluctuations. The development of this framework will have a positive impact on the domestic soybean value chain by developing agronomic solutions that support yield increases under varying environmental conditions using a holistic soil-crop-atmospheric systems modeling approach.

Review Of Relevant Literature

Uncertain impacts of year-to-year weather fluctuations on soybean yields and systems' water use efficiency

Agronomic advances, crop improvement, and weather shifts occur continuously, but their combined impacts on soybean system performance are not incremental (Iqbal et al., 2018). Year-to-year weather fluctuations are high, and crop management must be dynamic to maximize yields for target environments and production systems. Properly predicting crop response to water availability and temperature is critical in designing effective interventions that maximize productivity, but existing crop models still lack sufficient physiological detail to properly predict impacts of extreme weather fluctuations (Boote et al., 2021; Nóia Júnior et al., 2025). Previous research has indicated valuable but inconsistent results of soybean yield response to weather shifts (Boote, 2011; Jin et al., 2017; Archontoulis et al., 2020; Yu et al., 2021; Kothari et al., 2022). These inconsistencies present a major barrier to identifying promising agronomic management solutions. Because many of these studies are conducted using modeling techniques, we suggest that the inconsistencies arise from the use of varying methodologies, types of models (statistical or process-based), assumptions, adaptive scenarios, baseline weather conditions, and a lack of physiological detail to account for extreme weather conditions.

For example, high temperature events during flowering and seed filling can negatively impact crop yields because of reduced seed number and decreased partitioning of assimilates to grain. Research has demonstrated that optimum soybean harvest index (the fraction of above-ground biomass allocated to seeds) at 26°C gradually declined as temperature increases, reaching zero (failure point) at approximately 40°C (Boote et al., 2005). Crop seasons that are warmer than

historical average conditions can shorten the soybean crop growth period due to accelerated development rates, leading to less biomass accumulation and yields (Elli et al., 2022). Lower biomass accumulation will have implications on seasonal water demand and system's water use efficiency - WUE (Condon et al., 2004). On a global scale, every 1°C temperature increase is estimated to reduce soybean seed yields by 1.3% (Lobell and Field (2007). Rising temperatures also increase evaporative demand, exacerbating water requirements in irrigated systems (Asseng et al., 2009). Additionally, excessive rain during spring may cause sowing delays and expose the crop to less favorable weather conditions, decreasing yield potential (Rattalino Edreira et al., 2020). These complex interactions require systems frameworks capable of quantifying tradeoffs and synergies to inform decision-making for adapting crop management in accordance with weather conditions. **Without a framework to help us understand the impacts of current year-to-year weather fluctuations on soybean production systems, we will be unable to design effective solutions for target environments.**

Nearly 90% of the US soybean production is rainfed, and drought reduces 5-60% of the soybean yield in the US every year (NASS, 2023). Where rain is scarcer, irrigation is a critical component of the agricultural economy. For example, approximately 80% of the soybean cropland area is currently irrigated in Arkansas (NASS, 2023). Farmers in the US Mid-South have been facing challenges with water availability for irrigation, particularly during critical soybean developmental stages like flowering and seed filling, and concerns regarding the long-term sustainability of irrigation are rising (Reba et al., 2017). Thus, there is always interest in exploring strategies to alleviate drought stress and enhance WUE. Quantifying the impacts of agronomic management on seasonal water use could lead to the conceptualization of strategies that can be readily adopted by farmers to manage irrigation more efficiently and deal with limited water conditions. WUE is among the most instrumental variables for the long-term sustainability of soybean production, but little progress has been made in developing genotypes expressing specific physiological traits for drought (Sinclair et al., 2017). One reason is the lack of fundamental understanding of key physiological mechanisms that confer drought tolerance in soybean genotypes.

Higher plant WUE can be achieved by either increasing plant photosynthetic efficiency, decreasing stomatal conductance (and plant transpiration), or a combination of both (Condon et al., 2004). Vapor pressure deficit (VPD), which increases at higher temperatures, is the driving force of plant transpiration (Pan et al., 2015). A better understanding of soybean transpiration responses to VPD would enhance our ability to design drought-tolerant genotypes by identifying fundamental physiological mechanisms underlying WUE. For example, Fletcher et al. (2007) found that the 'slow-wilting' genotype PI 416937 could limit transpiration response under high VPD conditions and labeled this response as a "limited-transpiration" trait. Identifying additional soybean accessions that express this trait for use in breeding programs could positively impact the development of superior varieties for environments with insufficient rainfall or limited irrigation water.

A cropping systems framework to identify adaptive solutions to year-to-year weather fluctuations

Drought and high-temperature impacts have been mostly evaluated using single-point measurements of crop yields at physiological maturity (e.g., Battisti et al., 2017; Yu et al., 2021), rather than the quantification of seasonal dynamics in crop biomass accumulation, crop water use and losses by runoff and drainage (e.g., Freitas et al., 2025). For example, quantifying daily

dynamics of soil moisture and crop water deficit during the critical period for seed number determination (R3-R6 stages, Monzon et al., 2021) can help identify agronomic strategies to increase growth rate during that period and increase seed yields. A critical barrier in designing solutions that account for factors beyond seed yields is that some variables (e.g., water and nutrient runoff) can be prohibitively expensive and impossible to measure at high temporal and spatial resolutions, and crop systems modeling can address these limitations. Without the ability to mathematically describe systems dynamics using a model, we cannot fully understand systems outcomes. Crop simulation models that consider weather, soil, and crop inputs to simulate various processes of the cropping system across timescales can enhance our understanding of complex interactions and allow conceptualization of weather-dependent solutions (Holzworth et al., 2018). **We propose integrating multi-scale data (e.g., leaf, plant, field, system) and multiple processes of the cropping systems (e.g., crop growth, light capture, evapotranspiration, water losses) into a unique systems modeling framework to identify more assertive agronomic management prescriptions.**

Some challenges have been overlooked in using crop models for quantifying crop response to weather shifts. First, little attention has been given to improving models towards reducing uncertainties in predicted impacts (e.g., Nóia Júnior et al., 2023) and has instead focused on quantifying weather impacts per se (e.g., Fernandes et al., 2022). This research is particularly important for cropping systems evaluations. For instance, high temperature impacts on biomass production have cascading effects on crop evapotranspiration, surface residue production, soil water, soil temperature, and residue decomposition. Reducing uncertainties in biomass production simulations will likely reduce uncertainties in evapotranspiration and water balance predictions. Second, there is a critical need for systems-level perspectives to guide bioengineering strategies for crop improvement (Matthews and Burgess, 2024). Yet, our ability to use dynamic crop growth models to evaluate hypothetical yield-enhancing plant traits for potential genetic improvement is highly limited by insufficient representation of physiological processes within crop models (Boote et al., 2021).

The Agricultural Production Systems sIMulator (APSIM) is an advanced cropping systems modeling framework (Holzworth et al., 2014) that contains modules for simulating soil water balance, surface residue, soil carbon, N cycling, crop evapotranspiration (Probert et al., 1998; Huth et al., 2010; Figueiredo Moura da Silva et al., 2025), and growth and development of various crops (Keating et al., 2003). The APSIM soybean model calculates biomass production based on intercepted daily solar radiation, radiation use efficiency (RUE), and environmental stressors such as water, CO₂, temperature, and nitrogen (Holzworth et al., 2014; Vanuytrecht and Thorburn, 2017). This approach has proven successful for predicting soybean growth and seed yields in the US (Archontoulis et al., 2020; Elli et al., 2022). However, RUE is an emergent outcome resulting from the expression of many genes (Parent and Tardieu, 2014) without a proper connection to a measurable leaf-level physiology process that could be of greater value for breeders (Boote et al., 2021). Rising nighttime temperatures threaten soybean seed yields by increasing respiration rates (Djanaguiraman et al., 2013). An enhanced model predicting photosynthesis and respiration separately in response to hourly weather fluctuations would better simulate the individual and combined effects of day (heat stress) and nighttime temperatures (Oberhuber and Edwards, 1993).

We propose to advance current soybean model capabilities by implementing new algorithms within APSIM that simulate leaf-level photosynthesis, respiration, and stomatal conductance

processes using the Diurnal Canopy Photosynthesis–Stomatal Conductance (DCaPST) model (Wu et al., 2017). The DCaPST model captures diurnal environmental variations, canopy attributes, and photosynthetic attributes and calculates instantaneous gross assimilation (A) of the sunlit and shaded foliage areas, integrating the results hourly and at the crop level (i.e., biomass production and yields) (Wu et al., 2017). The DCaPST model routine has been successfully implemented in other crops within APSIM, such as sorghum, maize, and wheat (Wu et al., 2023, 2019), so implementing this routine into the soybean model will be a straightforward step. Having the DCaPST model implemented into APSIM-Soybean will allow us to better quantify the impacts of drought and heat on soybean growth dynamics and, therefore, increase accuracy in prescriptions to maximize soybean yields and water use efficiency.

Preliminary Data: experiments and modeling

We have collected new transpiration data in response to VPD levels (**Figure 1**) from unexplored maturity group (MG) IV soybean accessions with associated differences in the environmental plasticity of carbon 13 ratio, a surrogate measure for WUE (Chamarthi et al., 2023). We grew 4 accessions in controlled-environment chambers within the University of Arkansas Division of Agriculture (UADA) facilities (results from 2 accessions showing contrasting transpiration responses to VPD are shown in **Figure 1**). Our preliminary data showed that WUE plasticity was related to the ability of some genotypes to limit their transpiration rates under high VPD (**Figure 1, upper panel**), while other genotypes with no associated WUE plasticity continued to increase transpiration under high VPD (**Figure 1, lower panel**). However, the limited transpiration trait was not identified for all genotypes with expected WUE plasticity, suggesting that other mechanisms control WUE for certain genotypes. Expanding the number of soybean accessions will be important to confirm our initial findings. Moreover, we learned that an additional “high” VPD treatment (>3.4 kPa) is needed as some genotypes started to alter their transpiration patterns at very high VPD levels (data not shown). Lastly, it is unknown how commercial soybean varieties would benefit from limiting transpiration under different weather scenarios.

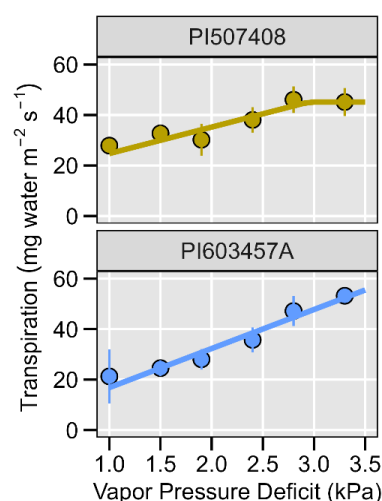


Figure 1. Transpiration responses of two contrasting soybean accessions (PI507408 and PI603457A) to vapor pressure deficit (VPD). Vertical lines represent the standard deviation across 4 replications.

This trait has been particularly desirable for limiting water use early in the growing season so that water is conserved for use later in the season, when it is needed to sustain plant growth rate during seed set and fill stages (Raymundo et al., 2023). However, restricting stomatal conductance under high VPD would limit carbon assimilation, which can decrease yield in some environments. Currently, the APSIM soybean model is not capable of simulating impacts of this trait. Once we have the DCaPST model routine implemented into the APSIM Soybean model, we will be able to quantify the value of the limited transpiration trait across different environments and identify in which geographical environments genotypes could benefit from expressing this trait as a strategy to enhance drought tolerance (Sinclair et al., 2010).

Current efforts of the APSIM initiative are transitioning the crop models to a new framework, the APSIM Next Generation (NG), which accounts for improvements related to execution speed, model construction and visualization, and manager script flexibility (Holzworth et al., 2018). We updated previous cultivar calibrations performed in APSIM Classic (Elli et al., 2022) to the APSIM NG soybean model and demonstrated the model's potential to capture physiological and yield differences in soybean cultivars of varying relative maturities in the US Midsouth (**Figure 2**). However, further testing is required across contrasting environments and production situations to increase confidence in modeling predictions.

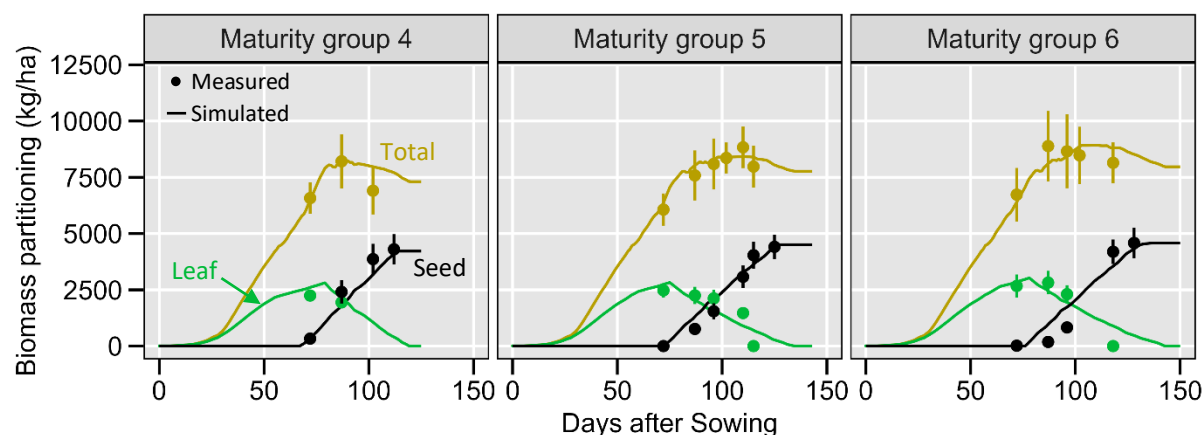


Figure 2. APSIM-simulated and measured leaf, seed, and total biomass in Northwest Arkansas. Measured data was obtained from Mastrodomenico and Purcell (2012) from an experiment conducted in a randomized complete design with 4 replications in Fayetteville, AR. Vertical lines represent the standard deviation across 4 replications.

We propose to use the APSIM soybean model in this project for the following reasons: a) the model has proven robust in simulating several cropping system variables across a range of production situations in US soybean systems, including soil and crop N dynamics, soil water dynamics, leaf area index, phenology, and yields (Pasley et al., 2020; Archontoulis et al., 2020; Elli et al., 2022); b) the current model version release (APSIM NG) has modern languages and frameworks that allow for much greater script flexibility providing an ideal environment for extending model capabilities to different levels of complexity (Brown et al., 2014); c) the model is used in over 120 publications per year, receives over 3500 citations per year, and its use in research is exponentially growing based on Web of Science metrics, and d) the PI has over 6 years of experience using and improving the APSIM framework (Elli et al., 2019; Elli and Archontoulis, 2023).

Our preliminary simulation exercise supports our central hypothesis that a system-level understanding of soybean yield and water use dynamics will facilitate the conceptualization of solutions to maximize seed yields and reduce farmer input under drought and high temperature risks (**Figure 3**). We ran the APSIM-Soybean model in a rainfed production situation in Northwest Arkansas across 74 weather years (1950-2024) to quantify the impacts of agronomic management and physiological traits on seed yield and the system's WUE. Weather information was retrieved from a local weather station linked to the National Weather Service network and classified into five classes based on season, temperature, and rainfall: cool/wet, warm/wet, cool/dry, warm/ dry, and "normal" conditions. "Normal" conditions followed the methodology

of Ruane and McDermid (2017). Two selected scenarios are presented in **Figure 3**. We determined systems' WUE using an approach similar to Dietzel et al. (2016), where $sWUE = \text{seed yields} / [\text{season evapotranspiration} + \text{annual runoff} + \text{drainage}]$. Our what-if scenarios considered genotypes with different maturity groups, early sowing, a hypothetical genotype with decreased maximum stomatal conductance, and increased RUE (Boote, 2011; Koester et al., 2014).

We considered a scenario that combines early sowing and using an early-maturing genotype (MG 3 instead of 4) as a strategy to avoid late-season drought and heat stress under warm/dry years. An early-maturing genotype could be a strategy for saving water over the growing season without compromising yields in certain production situations (Edwards et al., 2003; Salmerón et al., 2016). Soybean has been planted earlier in Arkansas by almost one day year⁻¹ over the last 30 years (NASS, 2023). Warmer springs likely allow for earlier planting if increased extreme rainfall events do not impact field workability (Huber et al., 2023).

Our simulation exercise indicated that crop management and physiological traits provided opportunities to enhance yields without increasing seasonal water demand. Earlier sowing combined with an early-maturing genotype sustained soybean seed yields in relation to the baseline condition in warm/dry years, while slightly increasing systems' WUE. The single strategy of using an early-maturing genotype decreased crop evapotranspiration, but penalized seed yields in both weather scenarios (**Figure 3**). Decreased maximum stomatal conductance in warm/dry years showed potential to increase yield, possibly due to saving soil moisture during vegetative growth for use during critical periods like seed filling (Sinclair, 2018). Increased RUE was more effective in increasing seed yield compared to the baseline in wet/cool years, without substantially increasing seasonal evapotranspiration. Our proposed task to implement the DCaPST model routine into APSIM will facilitate the exploration opportunities to increase seed yield by enhancing photosynthesis rather than RUE, given that RUE is an outcome of multiple physiological processes (Boote et al., 2021).

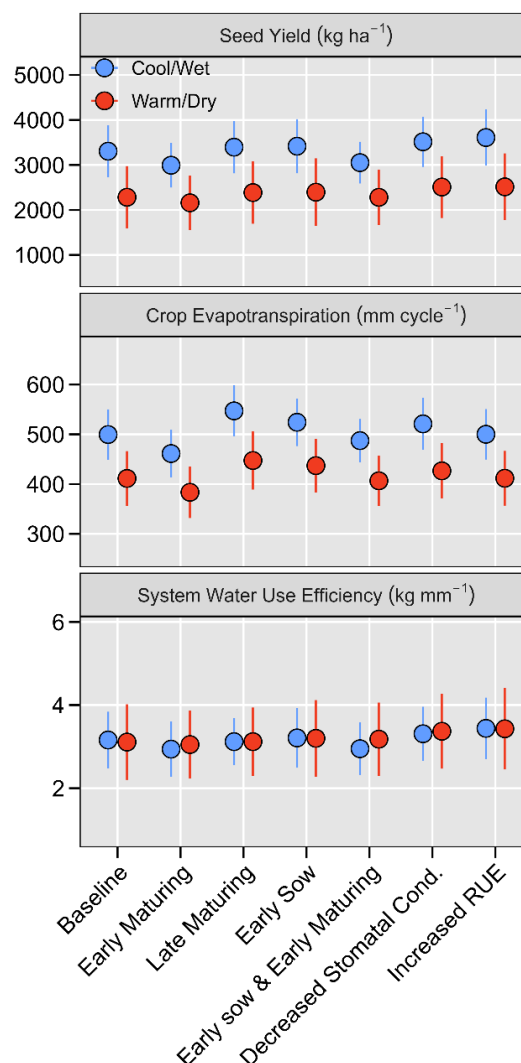


Figure 3. What-if scenario analysis to inform solutions in the context of year-to-year weather fluctuations. We used the APSIM-Soybean model version 2024.5.7502.0 to run a long-term simulation using 74 weather years and a soybean MG 4 variety (baseline) in Northwest Arkansas under rainfed conditions. Circles indicate the average, while vertical lines indicate the standard deviation across the 74 years.

Our preliminary results demonstrate: 1) the value of using a cropping system model combined with experimentation to understand dynamics that occur among management components of a production system, 2) the need for a better representation of physiological processes in the models to better capture crop responses to drought and high-temperature events, and 3) the need to create a systems framework to conceptualize alternative strategies that prevent soybean yield losses under extreme weather. We currently do not have a well-established and validated simulation modeling framework to inform decision-making in US Mid-South agriculture, which hampers our ability to advance research on how weather affects decision-making in soybean production systems.

Our project is novel for the following reasons: First, we propose a cross-scale physiological approach (leaf → plant → crop → system → region) that combines controlled-environment, on-farm, and crop modeling experiments to quantify impacts of extreme weather on soybean cropping systems' yield and seasonal water use. Second, this work represents a pioneering attempt to parameterize and implement the DCaPST model into a leguminous crop, a very needed work to enhance the level of physiological detail in quantifying impacts of drought and high temperatures on soybean yields. Third, our preliminary data reveal novel evidence of genotype response to VPD being associated with shifts in environmental plasticity in soybean WUE (Chamarthi et al., 2023). Lastly, our system's approach advances existing frameworks by mechanistically accounting for measurements beyond final seed yields, which is key to understanding system tradeoffs and improving agronomic decisions to cope with extreme weather events.

RATIONALE AND SIGNIFICANCE

Rationale

New knowledge of how various management components and physiological traits interact and impact soybean seed yield, systems water use and efficiency under variable weather conditions will lead to the conceptualization of innovative, sustainable solutions for preventing yield losses and increasing profitability in soybean production systems in the context of year-to-year weather variability, which is a factor that threatens soybean production every year in the US.

Our proposal addresses program Area A1102, “**Foundational Knowledge of Agricultural Production Systems**” by advancing knowledge of agricultural production systems and addressing critical dynamics that occur among management components of a production system using experimental manipulations of system components, technological interventions, system analyses, and modeling. Our proposal addresses two key priority requirements: “Investigate how changes to cropping systems affect crop performance and other outcomes beneficial to cropping systems” by testing how agronomic management and physiological traits can positively impact soybean systems performance using a cropping system model, and “Conduct syntheses and meta-analyses of existing data or develop new or extend existing models to derive general principles about the function, properties, and performance of agricultural production systems” by extending capabilities of a cropping system simulation model to advance understanding of weather fluctuations on soybean production systems. Our unique contribution is enhancing model capabilities and implementing them within a cross-scale systems framework to provide stakeholders with better prescriptions that help prevent yield losses under varying weather conditions.

Significance

Heat and drought events can decrease soybean seed yields by 19% in the US if no concurrent adaptive strategies are implemented (Elli et al., 2022). For the state of Arkansas, where soybean is the number one crop with approximately 3 million acres harvested in 2023 (NASS 2023), that would represent a yield loss of ~ 31 million bushels and ~ \$450 million annually to growers (Arkansas Agriculture Profile, 2023). High temperatures can drive lower crop biomass accumulation due to a faster development rate (Elli et al., 2022), increase the risk of yield failures (Hatfield and Prueger, 2015), and further intensify drought stress (Lobell and Gourdji, 2012). Consequences of extreme weather are particularly important for regions that already experience hot summers (frequent days with maximum temperatures above 40 °C) and late-season drought events, like the US Mid-South, challenging efforts to maintain and improve soybean production. Despite the growing magnitude of this problem, actionable strategies for preventing soybean yield losses due to emerging abiotic challenges that limit crop productivity like site-specific drought and high temperatures remain unclear. Outcomes from this proposal will generate new knowledge of how multiple management and physiological components of soybean cropping systems interact, which can help explain how seed yield and systems' water use can be manipulated by altering agronomic and cultivar settings. *This contribution is significant because this new knowledge will directly inform initiatives and recommendations to aid farmers, extensionists, and breeders in their attempts to mitigate the negative impacts of weather extremes on soybean cropping systems.* The generation of system-specific management recommendations for soybean producers is expected to maximize yields by informing dynamic decision-making in the context of local weather, which farmers can rapidly and realistically adopt.

3. APPROACH

The proposed approach involves two fundamental steps: 1) implementing new algorithms within a cropping system model to enhance predictability and explanatory power of physiological processes under drought and heat stress conditions, and 2) using a well-tested version of a cropping system model to identify the most effective solutions for preventing soybean yield losses while potentially maximizing seasonal water use across environments.

Objective #1: Implement new algorithms within a cropping system model (Agricultural Production Systems Simulator - APSIM) that operate at the leaf level and further test the model's ability to simulate soybean growth, seed yield, and water use.

Introduction. Further increases in crop yields will likely need to come from improving resource use efficiency (e.g., light, water), which is closely linked to photosynthetic efficiency (Long et al., 2015; Wu et al., 2019). A fundamental problem is that yield alterations caused by photosynthetic manipulation are commonly complicated by tradeoffs arising from interactions among genotype, environment, and management practices. A cross-scale modeling framework capable of linking leaf-plant-crop-level processes is very much needed to quantify and understand these tradeoffs. Presently, the APSIM soybean model lacks the capacity to explore leaf-level photosynthetic processes. Here, our goal is to implement and validate new equations to simulate leaf-level physiological processes within the APSIM cropping system model. Our working hypothesis is that the incorporation of new algorithms to account for more realistic trait physiology will improve the model's predictive performance and its capacity to explain key

physiological processes (Boote et al., 2021). We will conduct a controlled-environment chamber experiment to generate new fundamental data on soybean leaf CO₂ assimilation in response to intercellular CO₂ concentration, light, and temperature. We will then use this new data to parameterize the DCaPST leaf-level model within the APSIM soybean cropping system model.

Once we have DCaPST parameterized with our high-quality data, we will be able to identify potential traits for enhanced soybean yields and WUE for target environments. The expected overall outcome from this objective is to generate a new, well-tested version of the APSIM cropping system model with enhanced capabilities to generate a deeper physiological understanding of processes underlying WUE for accelerating crop improvement. This result is expected to have a significant impact on the field of agriculture because we will be able to design more robust strategies to maximize soybean systems' performance and water use efficiency under challenging environments.

Task 1.1: Generate new photosynthetic, stomatal conductance, and transpiration data. In controlled-environment chambers, we will grow 8 maturity group IV soybean accessions chosen to represent a spectrum of associated WUE differences based on the findings of Chamarthi et al. (2023). Soybean genotypes will be planted in 10-cm diameter, 40-cm long pots constructed of polyvinyl chloride pipe, sealed at the bottom and top with end caps and with a 2-cm diameter hole in the top through which the plant shoot will emerge and through which water will be applied (King and Purcell, 2017). The growth chamber will be maintained at day/night temperatures of 30/25°C (Yamazaki and Kawasaki, 2023) with a light cycle from 06:00 to 22:00 h. Photosynthetically active radiation at the top of the plants will be set to approximately 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ to avoid photosynthetic activity being under light-limiting conditions (Yao et al., 2017). Relative humidity will be set to 60% (King and Purcell, 2017). Leaf-level gas-exchange measurements will be taken at the beginning bloom (R1) crop stage on the central leaflet of the first fully expanded trifoliate from the top (Sakoda et al., 2019) to ensure the evaluated leaves receive direct sunlight.

The responses of CO₂ assimilation (A) to intercellular CO₂ partial pressure (A/Ci) and light (A/I) at different temperatures will be measured using a LI-6800 Portable Photosynthesis System (LI-COR Biosciences, Lincoln, NE, USA). Stomatal conductance will be determined using an LI-600 porometer (Li-Cor Biosciences, Lincoln, NE, USA). The experiment will be organized into two rounds, each containing 4 soybean accessions, to ensure timely data collection. The experiment will be laid out in a completely randomized design with 4 soybean accessions and four replications, totaling 16 individual plants. We will test temperature levels ranging from 15°C to 40°C with five °C intervals, CO₂ partial pressure from 50 to 1200 ppm using 24 steps at 3-min intervals, and solar radiation from 0 to 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ using 15 steps at 3-min intervals (Saathoff and Welles, 2021; Wu et al., 2024). For this particular task, running a controlled environment experiment instead of a field experiment will be critical to minimize external variability and ensure experiment reproducibility. Data analysis will be performed using R, where we will fit response curves to derive fundamental leaf-level physiological parameters (Wu et al., 2017).

Subsequently, plant-level transpiration responses to VPD will be evaluated using an approach similar to Fletcher et al. (2007). Briefly, individual pot tare weights will be recorded to monitor the pot mixture water content at the beginning of the experiment. Plants will be well-watered until the measurement date (beginning flowering, R1) by watering to 70% of pot-capacity weight as needed to keep soil water contents from falling below 50% of the pot-capacity weight (King et al., 2009). Portable balances will be installed under each pot with soybean plants (**Figure 4**). The balances will be linked to an Arduino microcontroller and datalogger system, which will measure and record individual pot weights every minute. When the VPD treatment is initiated, the pot weight between subsequent weighing times is assumed to be water lost through transpiration. Measurements will start at the lowest VPD treatment (1 kPa) to the highest (4 kPa) in 0.5 kPa increments (Fletcher et al., 2007) with a total of 6 VPD treatments.



Figure 4. Soybean plants grown on portable balances to monitor real-time transpiration rates.

Whole-plant transpiration rates will be normalized per unit leaf area ($\text{mg H}_2\text{O m}^{-2} \text{ s}^{-1}$). After transpiration measurements, the plant leaf area will be determined destructively by passing the leaves through a LI-3000C Portable Leaf Area Meter (Li-Cor Biosciences, Lincoln, NE, USA). Transpiration rates for each genotype will be regressed against VPD levels (Fletcher et al., 2007). Linear and non-linear (e.g., linear-plateau) regression models will be evaluated in R (R Development Core Team, 2021) using the packages *lm*, *lme4* (Bates et al., 2014), and *car* (Fox and Weisberg, 2019), and best-fit models will be selected based on lower Akaike Information Criterion (AIC) and the highest R^2 .

Task 1.2: Measuring on-farm soybean seasonal crop evapotranspiration and crop growth dynamics. In 2 commercial production fields in Lonoke County, Eastern Arkansas (in a farm where Runkle has worked previously, in Moreno Garcia et al. 2021), under irrigated conditions, field-level evapotranspiration (ET) will be measured in years 1 and 2 of the proposal using two “water node” evapotranspiration sensors (LI-710; LI-COR Biosciences). The 2 on-farm fields will be selected to capture variability in soil type, planting date, and genotype maturity group. The evapotranspiration sensors combine vertical wind speed and relative humidity measurements at high frequency to determine the ET rate. The sensors will be installed above the canopy of fields that are at least 10 ha in size to ensure accurate fetch to characterize field-scale ET. The sensors will operate continuously from sowing to physiological maturity, and flux values will be recorded at 15-minute intervals. These measurements will be supplemented by continuous measurements of soil moisture and temperature at 3 depths (15, 30, and 45 cm, covering the zone where fine roots are concentrated), weekly plant height, stage, and leaf area index (via Licor 2200 Plant Canopy Analyzer in the Runkle lab). While use of a relatively new sensor may produce some risk, Runkle’s group already has experience (via collaboration) managing and interpreting data from Licor’s similar “carbon node” instrument and found that it agreed closely with co-located eddy covariance datasets. His team also has ample experience with different micrometeorological techniques, such as the surface renewal and eddy covariance methods (Runkle et al., 2012, 2014; Simpson et al., 2019; Suvočarev et al., 2019; Fong et al., 2020; Castellví et al., 2020a; b).

An additional soybean field experiment will be conducted in years 1 and 2 of the proposal under rainfed and irrigated conditions at the UADA Research & Extension Center in Fayetteville, Northwest Arkansas, where the PI/Co-PI are housed. The goal with this experiment is to collect detailed, in-season crop growth data. The experiment will be conducted as a two-factor (genotypes and irrigation regime) factorial, which will be laid out in a randomized complete block design with four replications. We will have 8 genotypes $\times$ 2 irrigation regimes $\times$ 4 replications = 64 plots per year. Individual plots will have 8 rows of 6 m and a row space of 91 cm, following the experimental planter setup of the research station. We will use 4 maturity group IV commercial genotypes well adapted for the studied region and 4 soybean accessions that will be strategically selected based upon the results of the controlled environment experiment to capture variability in transpiration response to VPD (**Task 1.1**). PI Elli has enough seed from the accessions for the proposed field experiment, as a result of seed increase from previous research efforts.

We will collect time-series above-ground biomass samples by hand harvesting all plants from a 1m² plot area and partitioning the biomass into different organs (green leaves, senesced leaves, stem, pod walls, and seeds). Samples will be dried at 65 °C for 5 to 7 days and weighed. Biomass collection will be performed from V2 (second trifoliate leaf) to R7 (physiological maturity) stages at bi-weekly intervals. The leaf area index will be determined using a LI-3000C Portable Leaf Area Meter (Li-Cor Biosciences, Lincoln, NE, USA). Phenology will be recorded twice a week. Final yields will be determined when plots reach harvest maturity (R8 stage) using a small-plot combine, and seed weights will be corrected to 130 g kg⁻¹ moisture content.

Task 1.3: Incorporate leaf-level algorithms into the model. Leaf-level CO₂ assimilation (A/Ci and A/I) response curves, stomatal conductance and whole-plant transpiration data generated from **task 1.1** will be used to derive relevant coefficients for the Diurnal Canopy Photosynthesis–Stomatal Conductance - DCaPST model parameterization (Wu et al., 2017). As a starting point, we will use the structure and input parameters from the C3 wheat DCaPST model (Wu et al., 2023; Wu et al., 2023), which has already been implemented into the APSIM framework. After the standalone DCaPST parameterization, the model will be implemented into the APSIM NG (Holzworth et al., 2018) soybean model. The new algorithms will be employed using the Plant Modelling Framework – PMF (Brown et al., 2014), a modular framework for creating models that represent plant components within APSIM. Our approach will be similar to that used by PI Elli to add new algorithms into the C3 Eucalyptus model within APSIM (Elli et al., 2020a; b). Dr. Wu, the developer of the DCaPST model (Wu et al., 2017), is committed to providing intellectual support for the implementation of the model into APSIM Soybean (see support letter).

Task 1.4: Update APSIM model calibration and validation. After implementing the new algorithms into the APSIM soybean model, we will calibrate the entire crop model using data already collected by the PI (see preliminary results section) and new crop growth and evapotranspiration data from **task 1.2**. We will retrieve weather information from satellite-based products such as NASA/POWER (Stackhouse et al., 2015) and CHIRPS (Funk et al., 2015), which have proven robust for use in crop modeling (Luetkemeier et al., 2018; Battisti et al., 2019). Soil inputs will be obtained by combining SSURGO (Soil Survey Staff, 2023) information and soil moisture measured data from **task 1.2**. Model calibration will consist of manually changing cultivar parameters within a realistic range until a suitable fit is obtained (Elli et al., 2020a; Irmak et al., 2024). Briefly, the first step will be adjusting phenology-related

parameters to determine growing degree days and sensitivity to photoperiod so that simulated dates of flowering and maturity are within $\pm 15\%$ of the observations. A second step will be adjusting other model parameters related to leaf area and biomass partitioning within plausible ranges ($\pm 20\%$) as needed to achieve a target normalized Root Mean Square Error – nRMSE of 20% or lower for final seed yields. Additional model performance metrics include coefficient of Determination – R^2 (target criteria is $R^2 \geq 0.8$ or greater for final seed yields), Willmott Agreement Index - d (Willmott et al., 1985, target criteria is $d \geq 0.8$), and Modelling Efficiency Index – E (Nash and Sutcliffe, 1970, target criteria is $E \geq 0.75$).

We will use an independent dataset from the literature (Elli et al., 2022) for model validation, which accounts for 10 sites with at least 2 weather-years per location, including one site in Arkansas, two in Kansas, three in Nebraska, three in Iowa, and one in South Dakota. The production situations span a temperature gradient from May to September of 18.7 to 23.8 °C, rainfall of 390 to 604 mm, rainfed (Iowa) and irrigated (all other sites) conditions, and soybean maturity groups from 1 (South Dakota) to 6 (Arkansas). The data includes time series on crop biomass, leaf area index, seed yield, N uptake, soil nitrates, and biological N fixation. The Arkansas location includes an MG 6 non-nodulating genotype, allowing us to further test APSIM's nitrogen fixation routine using field data. An additional model validation will be conducted using historical yield data (1990-2024) from USDA-NASS (NASS 2023) to assess APSIM's performance in reproducing historical yield variability across the studied region.

Expected results. New CO_2 assimilation (A/C_i and A/I) response curves and whole-plant transpiration data can be used not only for the DCaPST model parameterization, but also to identify novel fundamental mechanisms underlying WUE and seed yields. The tested accessions will generate new transpiration versus VPD response curves like those shown in **Figure 1**. This dataset has the potential to identify genotypes with an enhanced ability to conserve water under stressed environments. Upon completion of Objective 1, we expect to have a well-tested version of the APSIM soybean model with enhanced capabilities to predict trait physiology. We expect the model to provide greater explanatory power of key physiological processes and less uncertainty in identifying adaptive solutions depending on weather conditions, and model performance in predicting final crop yields and biomass production to remain as robust as the predictions from the previous model version, as shown in **Figure 2**.

Potential Pitfalls and Alternative Strategies. Failure in data recording with the Arduino microcontroller and datalogger system could compromise whole-plant transpiration data. Constant monitoring and backup will minimize data loss risk. Having the experiment in growth chambers offers the flexibility of rerunning the experiment in a relatively short time frame (e.g., ~ 45 days). The on-farm experiments will be located on the East side of Arkansas, around 350 miles from Fayetteville, where the PI/Co-PI are housed. The Co-PI Runkle has already established successful experiments at those farms. Having the experiment from where we will collect in-season biomass in the Northwest location (Fayetteville, where the PI/Co-PI are housed) will help us collect the proposed measurements in a timely manner. Unanticipated weather events (e.g., storms) could compromise data collection from field experiments (though they will also enable us to measure and observe responses during these stress events). Having the experiment at three locations and two years will provide more flexibility to implement contingency plans and redirect resources to other sites if a given site experiences severe damage. A drawback of having a more detailed leaf-level model is the cost (computational power and time) of the extra parameters that will be introduced. Through sensitivity analysis, we will

quantify the relative importance of individual parameters to guide the model user and direct resources to the most influential parameters.

Objective #2: Identify the most effective management and physiological strategies for target environments that will increase soybean seed yield and water use efficiency.

Introduction. Designing solutions that account for factors beyond seed yields and that consider multiple aspects of the cropping system at a high spatiotemporal resolution is impossible without predictive models. Our goal is to implement a system framework to identify management practices and physiological traits that can increase soybean yields in the context of current weather fluctuations without compromising environmental performance. Given the dynamic and significant genotype, environment, and management interactions, our working hypothesis is that system-specific agronomic recommendations will maximize seed yields and water use efficiency. Therefore, we will use an improved and validated version of a cropping system model to run a what-if scenario analysis to identify sustainable solutions for adapting soybean systems to local weather fluctuations (e.g., **Figure 2**). New data generated from our crop physiology-modeling framework will lead to conceptualizing system-specific solutions that provide short-(agronomic management) and long-term (breeding) recommendations for preventing soybean yield losses under extreme weather. Upon completion of this specific objective, our expected outcome is to advance knowledge of how alterations to part of the system (management and physiological traits) will function in the context of year-to-year weather fluctuations. Filling this knowledge gap can contribute to increasing yields while reducing farm inputs in soybean cropping systems.

Task 2.1: Set up the APSIM model and generate extreme weather scenarios. Our simulation experiment will include ten environments in Arkansas (**Figure 5**). The selected environments are strategically spread across the state to cover a representative environmental gradient and relevant soybean production areas. Of the 10 sites, 8 have weather station capabilities belonging to the UADA research stations network, but not a complete dataset with historical time-series data necessary for crop modeling. Similar to task 1.4, we will retrieve weather information from NASA/POWER (Stackhouse et al., 2015) and CHIRPS (Funk et al., 2015), and soil information from SSURGO (Soil Survey Staff, 2023). A 10-year spin-up period will be run to reduce uncertainties regarding initial soil conditions (Dietzel et al., 2016). Row spacing and plant density inputs will follow the recommendations adopted by the University of Arkansas Cooperative Extension Service. APSIM-Soybean model is already set up for one location in Northwest Arkansas (**Figures 2 and 3**).

The scenario analyses will be evaluated using historical weather conditions to account for year-to-year weather fluctuations (**Figure 6**). Weather conditions will be characterized by grouping weather years into five classes based on season temperature and rainfall: cool/wet, warm/wet, cool/dry, warm/ dry, and “normal” conditions. “Normal” conditions include years in which weather conditions are within ± 0.5 * standard deviation of the long-term average values (Ruane and McDermid, 2017). The weather classification will allow us to dissect the impact of high

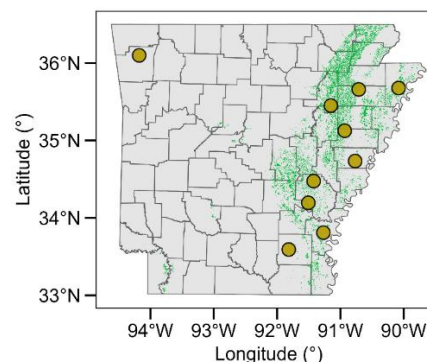


Figure 5. Arkansas map with the locations that will be used for the scenario analysis. The green background color denotes the 2023 soybean crop layer.

temperatures and drought events within realistic, historically observed ranges across the studied sites.

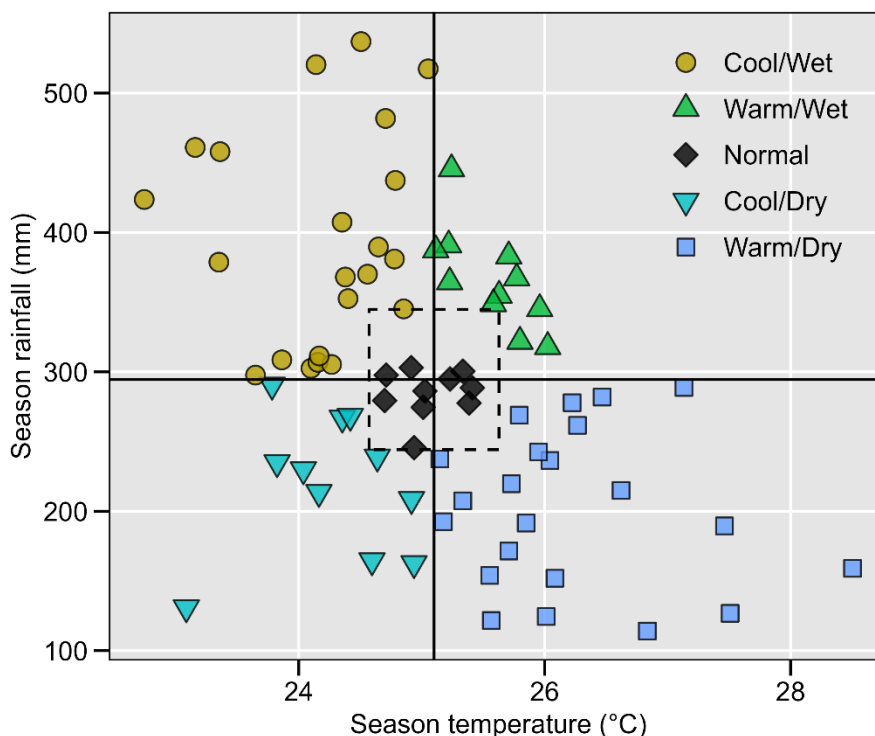


Figure 6. Classification of current year-to-year weather fluctuations for Fayetteville, AR (1950-2020). The dashed square in panel A represents $\pm 0.5 \times$ standard deviation of the long-term average.

Task. 2.2: Identify adaptive solutions through scenario analysis. Table 1 shows the adaptive scenarios to be explored and the metrics that will be generated to evaluate the system's performance and identify the most promising solutions. Management scenarios will be defined within realistic ranges identified in the literature (e.g., Mourtzinis et al., 2019) and common farmer practices adopted across the target research locations. Baseline information on sowing dates will be based on the USDA-NASS 50% state sowing progress (NASS 2022). We will run APSIM simulations over at least 40 historical weather years (1985-2024) to account for year-to-year weather variability. Initial model conditions will be reset every year on January 1 to avoid confounding signals from weather vs. nutrients on modeling outcomes (Elli and Archontoulis, 2023). We will identify optimum strategies (see below) for short- and long-term goals as we run the historical weather data scenario analysis. While the examples in Table 1 show only a few stacked options, additional combinations can be easily implemented in the scenario analysis. The DCaPST model, when implemented into the soybean model, will provide more flexibility in evaluating the hypothetical improvement of photosynthesis-related traits and the limited transpiration trait (Raymundo et al., 2023).

The dataset generated from the scenario analysis will be analyzed using linear models in R, where we will identify the optimum solution that maximizes yield and water use efficiency per environment. An R-shiny application (Chang et al., 2024) will be developed to create a user-

friendly web interface where users can visualize results together with optimum management and physiological solutions. Our goal with the web interface is to facilitate communication of the results with stakeholders, including private and industrial agriculture sectors, government organizations, and academics. We will add results from the what-if scenarios to help users understand the implications of different decisions on soybean yields and sustainability metrics (**Table 1**) according to weather scenarios (**Figure 6**). Outcomes from historical weather simulations (1990-2024) will influence actionable decision-making, which can be rapidly adopted by stakeholders. The R-shiny application will be web-based, ensuring accessibility through any standard web browser. Documentation will be provided to facilitate the use and interpretation of results. We will communicate with private farmers where the on-farm trials will be conducted (**task 1.2**) to share results from the simulation study through the web tool to improve farm management decisions and discuss practical ways to make the web tool more accessible to other stakeholders.

Table 1. Adaptive scenarios and systems variables to be generated using the APSIM model.

Adaptive scenarios
Management M1) early sowing date, M2) late sowing date, M3) early maturity group, M4) late maturity group, M5) no-till system, M6) full-till system, M7) more crop residue, M8) less crop residue, M9) high plant density, M10) low plant density, M11) more soil leftover N, M12) Less soil leftover N. Physiological traits G1) Enhanced photosynthesis (Boote, 2011), G2) limited transpiration under high VPD (Fletcher et al., 2007, Figure 2), G3) longer seed-filling period (Boote, 2011), G4) prolonged nitrogen fixation during drought (Devi and Sinclair, 2013), G5) slower senescence (Frederick et al., 1991). Stacked strategies (some examples) Early sowing date + early maturity group (S1), S1 + longer seed-filling period (S2), S2 + M1 + limited transpiration under high VPD.
Output variables/metrics to evaluate systems' performance in the scenario analysis
Yield levels and year-to-year variability, systems water use efficiency (Dietzel et al., 2016, $sWUE = \text{seed yields} / [\text{evapotranspiration} + \text{runoff} + \text{drainage}]$), cumulative intercepted radiation (Edwards et al., 2005), systems evapotranspiration, soil moisture at 15, 30 and 45 cm, daily crop growth rates.

Expected Results. The scenario analysis will generate a dataset that includes the adaptive scenarios, environments, years, and output metrics (**Table 1**). Upon completion of data analysis, we will identify optimal solutions that maximize farmers' yields across environments.

Potential Pitfalls and Alternative Strategies. What-if scenarios will generate a substantial number of simulations and a large dataset, which will require computational power to create and analyze the results in the proposed timeline. If needed, the Arkansas High-Performance

Computing Center (AHPCC) provides high-performance computing hardware, storage, and support services, including training and education to enable computationally intensive research at the university and within the state and collaborators elsewhere.

Dissemination of Results and Future Directions

We will disseminate our findings through peer-reviewed papers and scientific conferences. We will also communicate the results to farmers, industry, and policymakers through the Agricultural Communications Services with the Arkansas Agricultural Experiment Station, and to scientists at the annual Tri-Society meeting. The R-shiny web interface will be used to maximize dissemination of our research findings to a wide range of stakeholders. Co-PI Runkle will develop a two-period lesson on APSIM model and its application in this project to incorporate into his cross-listed undergraduate-graduate level course, Modeling Environmental Biophysics. PI Elli will disseminate results from this proposal into his graduate-level Crop Physiology course and undergraduate-level Soybean Production course. Students will use the proposed web-based tool to interact with the generated data, visualize prescriptions that maximize crop yields and water use efficiency. Our scenario analysis will consider 10 environments in the US Mid-South. Performing simulations at national scale is a logical next step but beyond the scope of this proposal. Our focus will be on soybean production systems in the US Mid-South, but the approaches implemented here can be extended to other soybean systems and environments as appropriate. Extending crop modeling efforts in the US Midsouth would leverage a range of future initiatives, including nutrient management, carbon cycling evaluations, precision agriculture applications, and integration with machine learning approaches to enhance genomic prediction accuracy.

Project Timetable

Study Objectives and Benchmarks	Year 1				Year 2				Year 3			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Objective #1												
Task 1.1: Generate new photosynthetic, stomatal conductance, and transpiration data												
Task. 1.2: Measuring on-farm soybean seasonal crop evapotranspiration and crop growth dynamics												
Task 1.3: Incorporate leaf-level algorithms into the model												
Task 1.4: Update APSIM model calibration and validation												
Objective #2												
Task 2.1: Set up the APSIM model and generate weather extreme scenarios												
Task. 2.2: Identify adaptive solutions through scenario analysis												